11 Double Integrals

Definition 11.1: Imprecise Definition of Double Integral

Let $f(x, y)$ be a function defined on a closed and bounded region $\mathcal{D}$ in the xy-plane. The **double integral** is defined by

$$\iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A$$

where $\mathrm{d}A$ is the area element in the xy-plane.

This integral can be interpreted as the volume under the surface $z = f(x, y)$ and above the region $\mathcal{D}$ in the xy-plane.

Below are some properties of double integrals that will be useful in evaluating them.

- $\iint_{\mathcal{D}} f(x, y) + g(x, y)\,\mathrm{d}A = \iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A + \iint_{\mathcal{D}} g(x, y)\,\mathrm{d}A$
- $\iint_{\mathcal{D}} c \cdot f(x, y)\,\mathrm{d}A = c \cdot \iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A$
- If $f(x, y) \leq g(x, y)$ for all $(x, y) \in \mathcal{D}$, then $\iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A \leq \iint_{\mathcal{D}} g(x, y)\,\mathrm{d}A$.

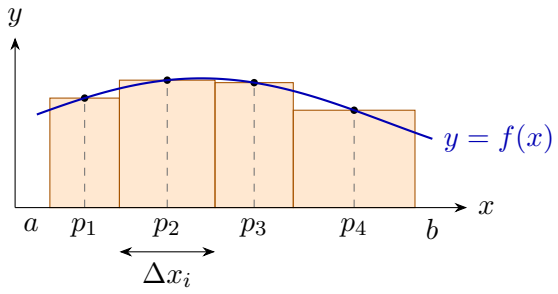
11.1 Riemann Sums and the Definition of Double Integrals

Recall the concept of Riemann sums for single-variable functions.

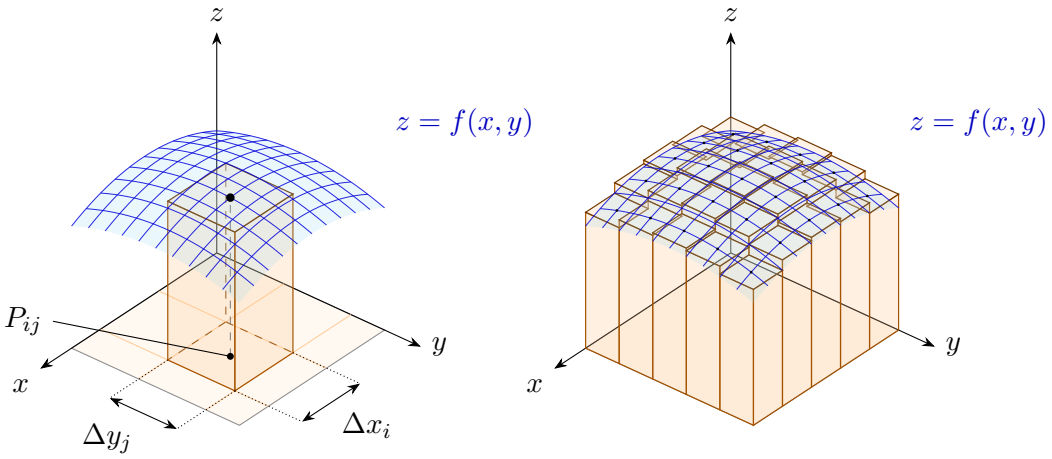
The Riemann sum for a function $f(x)$ over an interval $[a, b]$ is given by

$$\sum_{i=1}^{n} f(p_i)\Delta x_i$$

where Δx_i is the width of the i-th subinterval, and p_i is a sample point in that subinterval, as indicated in the figure below.



We can extend this idea to functions of two variables. Assume that $\mathcal{D} = [a, b] \times [c, d]$ is rectangular region in the xy-plane.



By this, we could write a precise definition of the double integral as follows:

Definition 11.2: Precise Definition of Double Integral

Let $f(x, y)$ be a function defined on a closed and bounded region $\mathcal{D}$ in the xy-plane. We partition $\mathcal{D}$ into m subintervals in the x-direction and n subintervals in the y-direction, forming a grid of rectangles.

Let Δx_i and Δy_j be the widths of the subintervals in the x and y directions, respectively, and let P_{ij} be a sample point in the rectangle formed by the i-th subinterval in the x-direction and the j-th subinterval in the y-direction.

The **double integral** of $f(x, y)$ over the region $\mathcal{D}$ is defined as the limit of the Riemann

sum as the maximum width of the subintervals approaches zero:

$$\iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A = \lim_{\|P\|\to 0} \sum_{i=1}^{m}\sum_{j=1}^{n} f(P_{ij})\Delta A_{ij}$$

Notice that the ΔA_{ij} here does not always exist. If it does, we call the function $f(x,y)$ **integrable** over the region $\mathcal{D}$. Also, if a function is continuous over a closed and bounded region, then it is integrable over that region.

11.2 Iterated Integrals and Fubini's Theorem

To evaluate double integrals, we can use Fubini's Theorem, which allows us to compute the double integral as an iterated integral.

Theorem 11.1: Fubini's Theorem

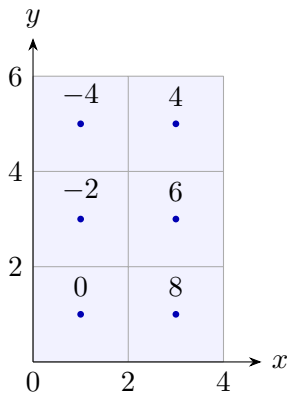
$$\iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A = \int_a^b \int_c^d f(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_c^d \int_a^b f(x,y)\,\mathrm{d}x\,\mathrm{d}y$$

Using this theorem, we can evaluate a double integral by computing two single-variable integrals in succession.

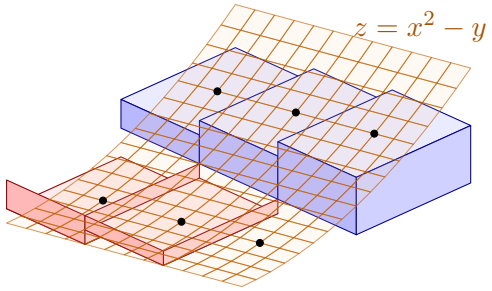
Example 11.1 (Evaluating a Double Integral)**.**

Let $f(x,y) = x^2 - y$ and $\mathcal{D} = [0,4] \times [0,6]$. We first approximate $\iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A$ using a midpoint Riemann sum. Divide $[0,4]$ into two equal intervals and $[0,6]$ into three equal intervals. This gives six rectangles, each with area $\Delta A = 2 \cdot 2 = 4$. The sample points are their centers:

$$(1,1),\ (1,3),\ (1,5),\ (3,1),\ (3,3),\ (3,5).$$



Midpoint samples, labeled by $f(x_i, y_i)$.



Midpoint prisms and the surface $z = x^2 - y$.

Thus the midpoint Riemann sum is

$$\begin{aligned} S &= \sum_{i=1}^{6} f(x_i, y_i)\Delta A \\ &= 4\big(0 - 2 - 4 + 8 + 6 + 4\big) = 48. \end{aligned}$$

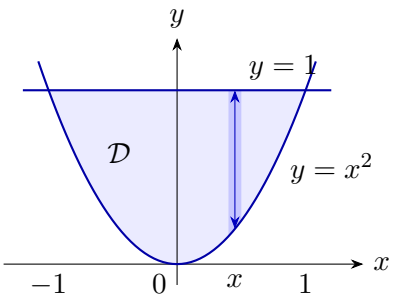
The negative terms represent contributions below the xy-plane.

By Fubini's theorem, the exact value is

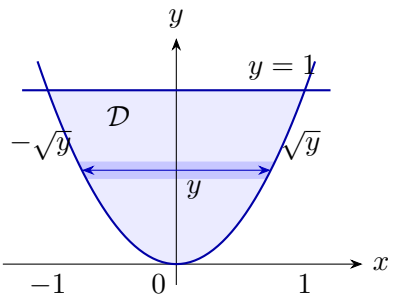
$$\begin{aligned} \iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A &= \int_0^4 \int_0^6 (x^2 - y)\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^4 \left[x^2 y - \frac{y^2}{2}\right]_0^6 \mathrm{d}x \\ &= \int_0^4 (6x^2 - 18)\,\mathrm{d}x \\ &= \big[2x^3 - 18x\big]_0^4 = \boxed{56}. \end{aligned}$$

For a non-rectangular region, the limits of the inner integral may depend on the outer variable.

Example 11.2 (Setting Up a Double Integral over a Non-Rectangular Region)**.**
Let $\mathcal{D}$ be the region bounded by $y = x^2$ and $y = 1$, and let f be continuous on $\mathcal{D}$. The curves intersect where $x^2 = 1$, so $x = \pm 1$.



Vertical slice: $x^2 \le y \le 1$.



Horizontal slice: $-\sqrt{y} \le x \le \sqrt{y}$.

For vertical slices, fix x and let y run from the lower boundary $y = x^2$ to the upper boundary $y = 1$. The slices cover the region as x runs from -1 to 1, so

$$\mathcal{D} = \{(x, y) \mid -1 \le x \le 1,\ x^2 \le y \le 1\}.$$

Hence

$$\iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A = \int_{-1}^{1}\int_{x^2}^{1} f(x, y)\,\mathrm{d}y\,\mathrm{d}x.$$

For horizontal slices, fix y and solve $x^2 \le y$ to obtain $-\sqrt{y} \le x \le \sqrt{y}$. The slices cover the region as y runs from 0 to 1, so

$$\mathcal{D} = \{(x, y) \mid 0 \le y \le 1,\ -\sqrt{y} \le x \le \sqrt{y}\}.$$

Thus the other order of integration is

$$\iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A = \int_{0}^{1}\int_{-\sqrt{y}}^{\sqrt{y}} f(x, y)\,\mathrm{d}x\,\mathrm{d}y.$$

11.3 Simple Regions

Definition 11.3: Simple Region

A region $\mathcal{D}$ is

- **Vertically simple** if

$$\mathcal{D} = \{(x, y) \mid a \le x \le b,\ g_1(x) \le y \le g_2(x)\},$$

where g_1 and g_2 are continuous on $[a, b]$ and $g_1(x) \leq g_2(x)$. Each vertical line intersects $\mathcal{D}$ in a single line segment, a point, or not at all.

- **Horizontally simple** if

$$\mathcal{D} = \{(x, y) \mid c \leq y \leq d,\ h_1(y) \leq x \leq h_2(y)\},$$

where h_1 and h_2 are continuous on $[c, d]$ and $h_1(y) \leq h_2(y)$. Each horizontal line intersects $\mathcal{D}$ in a single line segment, a point, or not at all.

If f is continuous on the corresponding simple region, then

$$\text{Vertically simple:} \quad \iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\,\mathrm{d}y\,\mathrm{d}x,$$

$$\text{Horizontally simple:} \quad \iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y)\,\mathrm{d}x\,\mathrm{d}y.$$

11.4 Average Value of a Function

Definition 11.4: Average Value of a Function

Let f be continuous on a closed and bounded region $\mathcal{D}$ in the xy-plane. The **average value** of f over $\mathcal{D}$ is defined by

$$\bar{f} = \frac{1}{A(\mathcal{D})} \iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A,$$

where $A(\mathcal{D})$ is the area of the region $\mathcal{D}$.

Problem 11.1: Average Value of a Function

Let $f(x, y) = x^2 + y^2$ and $\mathcal{D}$ be the region bounded by $y = x^2$ and $y = 1$. Find the average value of f over $\mathcal{D}$.

Solution 11.1

By definition, the average value is

$$\bar{f} = \frac{1}{A(\mathcal{D})} \iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A.$$

The area of $\mathcal{D}$ is

$$\begin{aligned} A(\mathcal{D}) &= \int_{-1}^{1}\int_{x^2}^{1} 1\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_{-1}^{1}(1-x^2)\,\mathrm{d}x = \left[x - \frac{x^3}{3}\right]_{-1}^{1} \\ &= \frac{4}{3}. \end{aligned}$$

Next, we compute the double integral:

$$\begin{aligned} \iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A &= \int_{-1}^{1}\int_{x^2}^{1}(x^2+y^2)\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_{-1}^{1}\left[x^2y + \frac{y^3}{3}\right]_{y=x^2}^{y=1}\mathrm{d}x \\ &= \int_{-1}^{1}\left(x^2 + \frac{1}{3} - x^4 - \frac{x^6}{3}\right)\mathrm{d}x \\ &= \left[\frac{x^3}{3} + \frac{x}{3} - \frac{x^5}{5} - \frac{x^7}{21}\right]_{-1}^{1} \\ &= 2\left(\frac{1}{3} + \frac{1}{3} - \frac{1}{5} - \frac{1}{21}\right) = 2\left(\frac{14+14-21-5}{105}\right) = 2\left(\frac{2}{15}\right) = \frac{4}{15}. \end{aligned}$$

Thus the average value is

$$\frac{1}{\frac{4}{3}} \cdot \frac{4}{15} = \frac{1}{5}.$$

Theorem 11.2: Linearity of Double Integrals

Let f be a continuous function on the xy-plane. If $\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2$, where $\mathcal{D}_1$ and $\mathcal{D}_2$

overlap only on their boundaries or not at all, then

$$\iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A = \iint_{\mathcal{D}_1} f(x,y)\,\mathrm{d}A + \iint_{\mathcal{D}_2} f(x,y)\,\mathrm{d}A.$$

By this theorem, we can compute the double integral over a complicated region by breaking it into simpler regions.